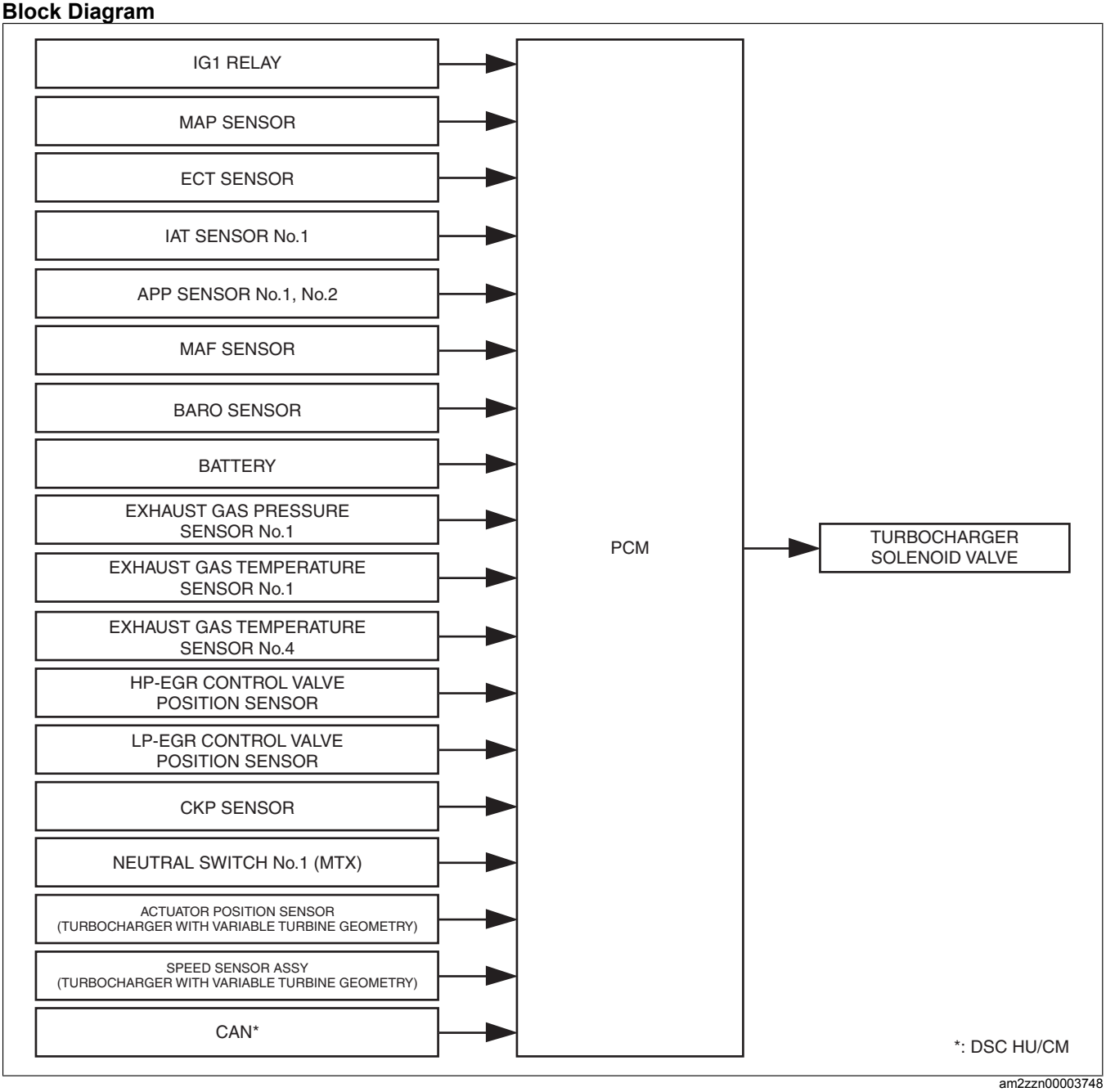
BOOST CONTROL [SKYACTIV-D 1.5]

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Outline

- The PCM controls the air charging pressure to improve the variable geometry turbocharger responsiveness, the acceleration performance and fuel economy of the vehicle.
- To obtain optimum air charging pressure according to the driving conditions, the PCM drives and controls the turbocharger solenoid valve.

Block Diagram



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Operation

- The PCM controls the turbocharger solenoid valve according to the engine operation conditions (engine speed and fuel injection amount) so that optimum air charging pressure is obtained.
- If driving conditions require increased air charging pressure, the duty value of the signal sent to the turbocharger solenoid valve is higher. At that time, the variable nozzle of the VG turbocharger moves in the closed direction.
- If driving conditions require decreased air charging pressure, the duty value of the signal sent to the turbocharger solenoid valve is lower. At that time, the variable nozzle of the VG turbocharger moves in the open direction.
- The PCM calculates the target air charging pressure of the variable nozzle based on the input signal and performs feedback control.

  Target lift of variable nozzle

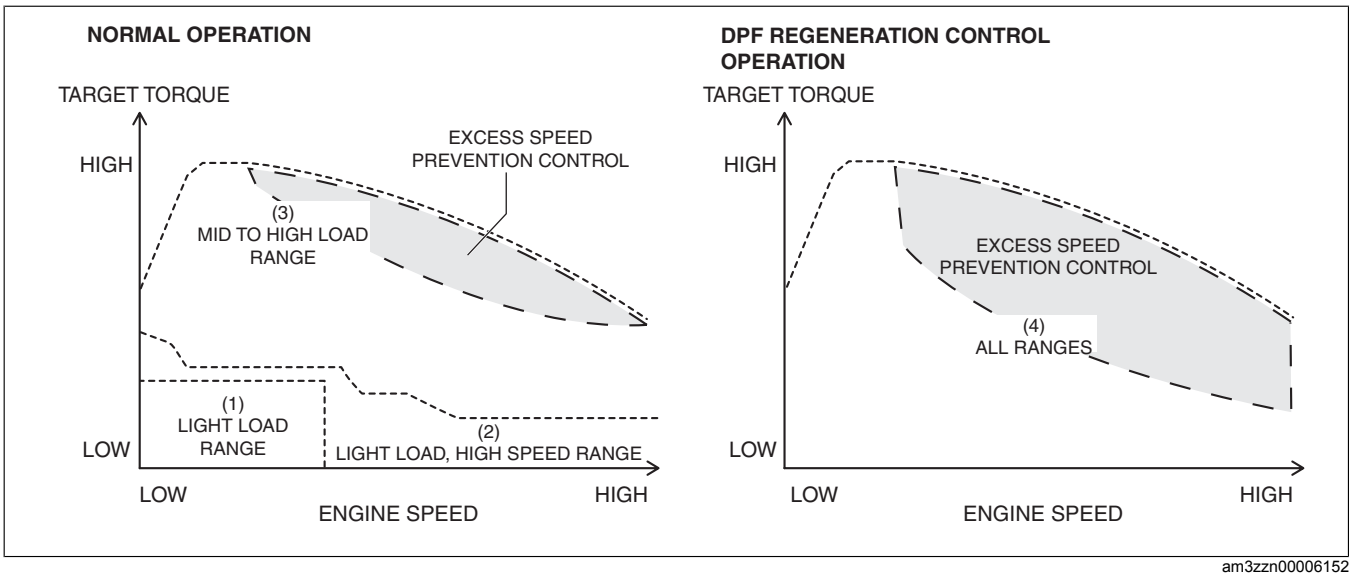
  — Target lift of variable nozzle is calculated based on the engine speed and target torque.

Target air charging pressure

— Target boost pressure is calculated based on the engine speed and target torque.

- The PCM adjusts the variable nozzle to the optimal opening angle in mainly 4 control ranges. In addition, the fuel injection amount is restricted according to the turbine speed to prevent the turbocharger with variable turbine geometry from operating in excess speed.

Control range (light load range to all ranges) image diagram



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—: Not applicable

Item		Turbocharger solenoid valve drive condition	Variable nozzle operation condition	Control status	Excess speed prevention control (fuel injection amount restriction)
	(1) Light load range	Duty 40—45%	Open	Variable nozzle lift amount fixed at 4.0 mm {0.16 in}*1	—
		Duty 0—80%	Open ⇔ Closed	Feedback control relative to target lift of variable nozzle	—
	(2) Light load, high speed range	Duty 0—80%	Open ⇔ Closed	Feedback control relative to target lift of variable nozzle	—
	(3) Mid to high load range	Duty 0—80%	Open ⇔ Closed	Feedback control relative to target lift of variable nozzle	Fuel injection amount is controlled when the turbine speed is 255,000 rpm or more.
		Duty 0—80%	Open ⇔ Closed	Feedback control relative to target air charging pressure*2	
	(4) All ranges (DPF regeneration control operation)	Duty 0—80%	Open ⇔ Closed	Feedback control relative to target air charging pressure	
Ignition is switched off		Duty 95%	Closed	Performs output error learning for the turbocharger with the fully closed position detected by the actuator position sensor as standard.	—
When engine is stopped by i-stop control		Duty 60%*3	Fully closed	Variable nozzle lift fixed at fully closed	—
No-load racing		Duty 0%	Fully open	Variable nozzle lift fixed at fully open	—

*1 : Only when HP-EGR control valve is open

*2 : When the change in the accelerator opening angle is small and the difference between the target air charging pressure value and the actual is within the specification.

*3 : By applying approx. 60% or more duty ratio sent to the turbocharger solenoid valve, the variable nozzle lift of the variable geometry turbocharger is fixed at fully closed.

- For details on the open/close of each valve in the air charging pressure control, refer to the “AIR CHARGING SYSTEM”. (See AIR CHARGING SYSTEM [SKYACTIV-D 1.5].)